Path Queries

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \ldots, n$, and node 1 is the root. Each node has a value.

Your task is to process following types of queries:

1. change the value of node s to x
2. calculate the sum of values on the path from the root to node s

Input

The first input line contains two integers n and q: the number of nodes and queries. The nodes are numbered $1, 2, \ldots, n$.

The next line has n integers $v_1, v_2, \ldots, v_n$: the value of each node.

Then there are $n-1$ lines describing the edges. Each line contains two integers a and b: there is an edge between nodes a and b.

Finally, there are q lines describing the queries. Each query is either of the form "1 s x" or "2 s".

Output

Print the answer to each query of type 2.

Constraints

- $1 \le n, q \le 2 \cdot 10^5$
- $1 \le a, b, s \le n$
- $1 \le v_i, x \le 10^9$

Example

Input

```
5 3
4 2 5 2 1
1 2
1 3
3 4
3 5
2 4
1 3 2
2 4
```

Output

```
11
8
```